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July 16, 2026

By electronic transmission

Ms. Tekedra Mawakana and Dr. Dmitri Dolgov
Co-Chief Executive Officers
Waymo LLC
Mountain View, California

Re: Exploratory inquiry — a physically fail-closed enforcement and accountability layer beneath the Waymo Driver

Dear Ms. Mawakana and Dr. Dolgov:

I am the founder of Ghost Autonomy, a pre-silicon research program based in Halifax, Canada, developing what we call an **enforcement-and-accountability layer for bounded autonomy**: a risk-monotone actuation boundary terminating in non-programmable analog hardware, an evidence-admissibility discipline applied *before* sensor fusion, and a constitutionally governed memory in which every change to a system's operating truth is a logged, challengeable event. The program's distinguishing practice is evidentiary: every claim in our corpus carries a graded standing — measured, proven, proposed, or notional — and no claim is permitted to exceed it. This letter observes the same rule.

Waymo's position needs no rehearsal from us: roughly half a million paid driverless rides per week, more than 220 million rider-only miles, and the only safety case in the field that is published, benchmarked against human baselines, and peer-reviewed. We write because of two things your program disclosed within the past eight months, which we read as defining its next phase. First, your December 2025 description of a separate onboard validation layer bounding what the learned planner may do — a public acknowledgment that scaled autonomy requires an explicit constraint layer, not only a better driver. Second, the events of May and June 2026 — the suspension of freeway service after construction-zone behavior, and the associated recall — which demonstrated that even the industry's deepest validation regime discovers unknown-unsafe scenarios in production, one software release at a time. Dr. Dolgov has said publicly that the binding constraint on expansion is validation rather than capability. We agree — and we believe that constraint is, at bottom, an assumptions problem.

At this point in your program's trajectory, we believe the most consequential engineering object you own is the set of assumptions on which it will continue — and that this set now deserves explicit registration, physical enforcement, and independent audit.

The offer. Ghost is prepared to pivot its development program to maximize compatibility with the Waymo Driver: our conformance instrument built to consume your scenario and trace formats; our admissibility layer parameterized to your multimodal suite — cameras, lidar, radar, and audio, exactly as fielded — and our guard-ECU interface work targeted at your vehicle platforms' actuation interfaces rather than a generic reference. Nothing in this would touch your

driver; the layer is designed to treat the intelligence above it as a fallible proposer, which is what any enforcement layer must assume.

What Ghost brings — specifically, and at its true standing.

1. **A measured boundary primitive.** An electrically isolated analog safety veto with ~ 32 ns intervention latency, witnessed on FPGA **[MEASURED]** — deliberately the only latency figure our corpus reports as measured rather than projected.
2. **Proven envelope mathematics.** The risk-monotone actuation algebra in its box case: projection is coordinatewise clipping; enforcement is firmly non-expansive (1-Lipschitz); the two-stage clamp is exact in the feasible case **[PROVEN]**.
3. **A fully specified refusal chain.** Five stages (S0–S4), from trusted numerics through fidelity gating, layered safety filtering, and risk-monotone envelopes to the analog veto, with six acceptance criteria (G1–G6) restated as a conjunctive, fail-closed conformance gate — published as criteria, not achievements **[PROPOSED]**.
4. **An admissibility-before-fusion evidence discipline.** Sensors as witnesses under integrity, alignment, scope, and lineage gates; correlated witnesses counted once; inadmissible evidence excluded rather than averaged in — sensor-agnostic by construction **[PROPOSED]**.
5. **A constitutional memory doctrine.** Append-only canon, anti-silent-drift amendment semantics, verification burden that increases with residence time, and recall as a priced contract (Governed Knowledge Systems trilogy, July 2026) **[NOTIONAL]**.
6. **A self-auditing corpus.** An 89-page consolidated foundational corpus (May 2026); a 94-page technical documentation edition, v3.0 (June 2026); and a 367-page compendium whose provenance ledgers grade every claim — including a published coverage map (68 subcategories: 21 full, 28 high, 9 partial, 10 gap) and a formal proof-debt ledger (P-1–P-7) naming what remains unproven.
7. **Empirical provenance, honestly scoped.** Physics-informed reduced-order kernels on elliptic-integral/AGM primitives with machine-precision convergence **[PROVEN]**, validated at $R^2 \approx 0.78\text{--}0.87$ on 200+ experimental datasets within their demonstrated envelope **[MEASURED]** — in a non-automotive domain, with automotive transfer registered in our own ledger as an open validation obligation.

What this could be worth to your program — specifically.

1. **A structural complement to your statistical case.** A conjunctive conformance gate bounds what a residual error *can do* — addressed to precisely the long-tail scenarios (construction zones, school-bus interactions) that statistical coverage reaches last.
2. **A constraint layer below the constraint layer.** Your validation layer is software within the programmable stack; ours terminates in electrically isolated, non-programmable hardware — an independent last resort that strengthens the safety narrative before federal reviewers.
3. **Machine-checkable assumption registries for ODD expansion.** Each new city, freeway class, or weather regime enters with its assumptions registered, enforced, and auditable — supporting the per-city validation economics behind your stated one-million-rides-per-week objective.
4. **Governed memory for foundation-model updates.** Anti-silent-drift semantics make every change to the distilled onboard models a logged, challengeable event — an artifact your December 2025 disclosure implies but does not yet name.

What Ghost is not. We have no fabricated silicon, no vehicle deployment, and no fleet data; our measured hardware evidence is presently a single number. We state this plainly because we regard the statement itself as an asset — a partner adopting our layer inherits a provenance ledger, not a marketing narrative — and because it is precisely why a compatibility-first alignment now, while our interfaces are still being fixed, costs little and shapes much.

Our inquiry is deliberately modest: *is such an alignment of interest?* As a first, zero-obligation step, we would run our conformance instrument against a trace or scenario format of your choosing and return the results with their standings — an artifact your team keeps regardless of what follows. Our two strategy volumes and the published corpus are available on request; deeper documentation under mutual NDA.

Respectfully,

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Evidentiary standings used in this letter — **[MEASURED]** witnessed on hardware; **[PROVEN]** a mathematical result; **[PROPOSED]** specified and defended, not yet validated; **[NOTIONAL]** a research direction, quotable as intent only. It is our house rule that no statement may exceed its standing; we extend the same rule to this letter.